

### Advanced (Habanero)

**Q1.** Solve for  $x$ :  $2^x = 8$

- (A)  $x = 2$
- (B)  $x = 3$
- (C)  $x = 1$
- (D)  $x = 4$
- (E)  $x = 0$

**Q2.** Solve for  $y$ :  $3^y = 27$

- (A)  $y = 1$
- (B)  $y = 2$
- (C)  $y = 3$
- (C)  $y = 4$
- (D)  $y = 0$

**Q3.** Solve for  $z$ :  $4^z = 16$

- (A)  $z = 1$
- (B)  $z = 2$
- (C)  $z = 3$
- (D)  $z = 4$
- (E)  $z = 0$

**Q4.** Solve for  $x$ :  $5^x = 25$

- (A)  $x = 1$
- (B)  $x = 2$
- (C)  $x = 3$
- (D)  $x = 4$
- (E)  $x = 0$

**Q5.** Solve for  $y$ :  $2^y = 32$

- (A)  $y = 3$
- (B)  $y = 4$
- (C)  $y = 5$
- (D)  $y = 6$
- (E)  $y = 1$

**Q6.** Solve for  $z$ :  $3^z = 9$

- (A)  $z = 1$
- (B)  $z = 2$
- (C)  $z = 3$
- (D)  $z = 4$
- (E)  $z = 0$

**Q7.** Solve for  $x$ :  $4^x = 64$

- (A)  $x = 1$
- (B)  $x = 2$
- (C)  $x = 3$
- (D)  $x = 4$
- (E)  $x = 0$

**Q8.** Solve for  $y$ :  $5^y = 125$

- (A)  $y = 1$
- (B)  $y = 2$
- (C)  $y = 3$
- (D)  $y = 4$
- (E)  $y = 0$

#### Open Ended Questions (FRQ)

**Q9.** Solve for  $x$ :  $2^x + 3^x = 35$

**Q10.** Solve for  $y$ :  $4^y - 2^y = 12$

#### Chef's Tips

- Read each question carefully.
- Use the space provided for free response questions.
- Check your work before submitting.